



Background & Motivation

Problem: Full fine-tuning of large pre-trained models (LLMs, ViTs, Wav2Vec) requires updating billions of parameters — prohibitive for most practitioners.

LoRA: Injects trainable low-rank matrices $\Delta W = BA$ into frozen layers. Efficient, but *couples* the magnitude and direction of weight updates.

DoRA Key Insight: Decompose each weight into a learnable **magnitude** scalar m and a **unit-norm direction**, then apply LoRA only to the direction:

$$W' = \frac{m}{\text{scale}} \cdot \frac{W_0 + BA}{\|W_0 + BA\|_{\text{col}}}$$

B initialised to **0**; m to column norms of W_0 . Overhead vs. LoRA: $+d$ scalars per layer ($< 0.1\%$).

Goals:

- ▶ Reproduce DoRA gains across four modalities
- ▶ Identify when magnitude decoupling matters
- ▶ Compare DoRA vs. LoRA at equal parameter budgets

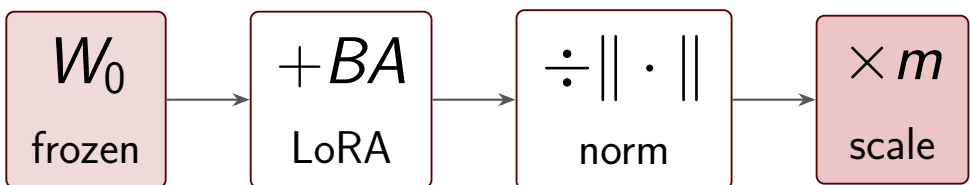
Methodology

Models & Tasks:

- ▶ **NLP:** RoBERTa-base/large on GLUE (SST-2, MRPC, RTE)
- ▶ **Audio:** Wav2Vec2-base on Speech Commands (12 KWS)
- ▶ **Vision:** ViT-B/16 & SigLIP-B/16 on robot grasping
- ▶ **VLA:** Action prediction on Push-T robot task

Shared Config: Rank $r \in \{2, 4, 8, 16, 32\}$; $\alpha = 2r$; target q,k,v,o.proj; AdamW + cosine LR; 5 epochs.

DoRA Decomposition Flow:



Implementation: Custom DoRALinear layer in PyTorch replacing every target nn.Linear in-place via `apply_dora_to_model()`. Adapter state (magnitude m , A , B) saved separately from frozen base weights.

NLP Results — GLUE Benchmark

MRPC Rank Sweep: (RoBERTa-base; combined F1 + accuracy)

Rank	DoRA (%)	LoRA (%)	Δ
2	87.1	86.2	+0.9
4	86.6	87.7	−1.1
8	87.8	85.9	+1.9
16	90.3	87.3	+3.0
32	89.8	90.1	−0.3

DoRA at $r = 16$ (90.3%) beats LoRA's best ($r = 32$: 90.1%) at **half the parameters**.

Cross-Task at $r = 8$: (RoBERTa-base accuracy / F1+acc / accuracy)

Task	DoRA	LoRA	Δ
SST-2 (67k, sentiment)	93.2%	93.3%	−0.1
MRPC (3.7k, paraphrase)	87.8%	85.9%	+1.9
RTE (2.5k, inference)	70.0%	66.1%	+3.9
SST-2 <i>large</i>	96.4%	95.5%	+0.9
MRPC <i>large</i>	89.3%	89.5%	−0.2
RTE <i>large</i>	76.9%	78.7%	−1.8

DoRA's advantage is largest on **low-data tasks** (RTE +3.9 pp) where magnitude decoupling reduces directional gradient interference. On large models both methods saturate.

Audio Results — Speech Commands

Wav2Vec2-base (67M params) fine-tuned on Google Speech Commands v0.02 — 84.8k utterances, **12 keyword classes**.

Method	Val Acc	Test Acc	Trainable	Time
DoRA ($r = 8$)	98.6%	89.7%	826.6k	152.8 min
LoRA ($r = 8$)	98.5%	89.0%	789.8k	190.8 min

+0.7% test accuracy, 25% less training time. The magnitude scalar's small overhead is offset by faster directional gradient convergence on the audio encoder.

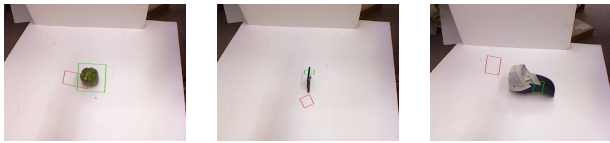
Vision & Robot Results

Object Grasping: RGB image $\rightarrow$ 2D bounding-box prediction using ViT-B/16 and SigLIP-B/16 backbones with DoRA/LoRA ($r = 8$).

Backbone	DoRA	LoRA
ViT-B/16	11.9% IoU	11.9% IoU
SigLIP-B/16	22.0% IoU	21.5% IoU

SigLIP's richer vision-language pretraining benefits more from magnitude decoupling. ViT features show minimal difference.

Qualitative: SigLIP + DoRA: Green = predicted grasp Red = ground truth



Push-T VLA Task: DoRA applied to a 126-layer Vision-Language-Action model for robotic push-block manipulation. Action MSE drops from 63.6 (epoch 1) to **27.9** (epoch 3).



DoRA adapter driving Push-T rollouts

Conclusions & References

- ▶ **Low-data NLP:** largest gains on RTE (+3.9 pp) and MRPC (+3.0 pp @ $r = 16$), where magnitude decoupling reduces gradient interference
- ▶ **Audio:** +0.7% test accuracy and 25% faster training on Wav2Vec2 + Speech Commands
- ▶ **Vision:** SigLIP benefits more than ViT; gains are modality- and architecture-dependent
- ▶ Gains **diminish on large models / easy tasks** where LoRA already saturates
- ▶ DoRA is a **drop-in LoRA replacement** across all four modalities tested

[1] Liu et al. (2024). DoRA: Weight-Decomposed Low-Rank Adaptation. *ICML 2024*.

[2] Hu et al. (2022). LoRA. *ICLR 2022*.

[3] Wang et al. (2018). GLUE Benchmark. *EMNLP*.

[4] Warden (2018). Speech Commands. *arXiv*.